Dissociating Kinematics, Force, and Control in Video World Models for Manipulation

Sangoh Lee[*]
GSAI, POSTECH
solee@dblab.postech.ac.kr

Sangwoo Mo[†]
IME, POSTECH
sangwoo.mo@postech.ac.kr

Wook-Shin Han[†]
GSAI, POSTECH
wshan@dblab.postech.ac.kr

Abstract

Project-status note (2026-04-30). This document is no longer a paper draft. It is a working record of the experimental state after the 2026-04-23 methodology reset. All previous abstract content was based on the episode-mean probing pipeline (`archive_old_wrong_probe/`) and is invalidated. The current artifacts are layer-wise probing results on the recollected PhysProbe dataset (12,100 episodes, valid contact GT) using V-JEPA 2 ViT-L raw residual features, plus a battery of latent-trajectory analyses. Sections that follow record *Setup / Outputs / Key observations / Known limitations / Status* per experiment block; no narrative or claim is intended.

Note 1 — Evidence inventory

Status. Replaces the prior introduction. All earlier claims (objective-specific dissociation, PEZ peak layers in the 11–12 range, native contact-force ranking V-JEPA 2 > VideoMAE-L > DINOv2-L, action-chunk control results) were produced by the invalidated episode-mean pipeline. They are not relied on here. The current valid evidence is restricted to V-JEPA 2 ViT-L (no VideoMAE / DINOv2 reruns yet), Variant A and Variant B feature pools, and the trajectory-analysis battery on the recollected dataset.

What we have right now.

- Layer-wise per-target Ridge probing across all six tasks under *Variant A* (1024-d spatiotemporal mean) and *Variant B* (8192-d, 8 temporal slots $\times$ 1024-d spatial mean).
- Extended target set per task: kinematic, rotational, contact, and progress-related variables (e.g. `insertion_depth`, `drawer_opening_extent`, `axial_progress`).
- Trajectory analysis on Variant A: per-layer intrinsic dimension, path/curvature/direct-distance ratios, cross-layer CKA, cross-task CKA, partial RSA controlling for pos+vel, tangent RSA, event-locked speed transients, kinematic-subspace CCA, Koopman self-predictability.
- Trajectory analysis on Variant B: per-temporal-slot decomposition, temporal-order ablation (7 schemes), cross-task transfer probing, phase/contact-conditional probing.

[*]Corresponding author.
[†]Equal advising.

Preprint.

- Episode-level physics-parameter probing (Variant A, 5-fold over episodes), reading per-episode invariants from `meta/episodes.jsonl`.

What we do *not* have. Cross-model comparison (VideoMAE, DINOv2), cross-scale (ViT-G), action-chunk control evaluation, and any benchmark-vs-baselines table are out of scope for this note. The fig1/fig2/.../fig5 figures in `Figures/` are stale and should be regenerated from current artifacts before reuse.

Reading order. Each subsequent section records one block: dataset and probe setup (Section); contact and force probing (Section); latent-trajectory geometry (Section); cross-task transfer and temporal-order ablation (Section); open questions (Section); artifact manifest and exclusions (Appendix).

Note 2 — Related work (deferred)

The prior related-work section was tightly coupled to the (now invalidated) narrative around objective-specific dissociation, native contact-force ranking, and OOD action regression. Until those claims are re-derived from the post-reset pipeline, the related-work discussion is intentionally omitted from this note. The bibliography in `references.bib` is preserved as-is for later reuse; entries are not pruned.

Note 3 — Dataset, features, probe setup

Setup

Dataset. Recollected PhysProbe (canonical path: `/home/solee/data/data/isaac_physrepa_v2_recollected_2026-04-23/`). LeRobot V2 format. Six tasks; 12,100 episodes total: Push (1,500), Strike (3,000), Reach (600), Drawer (2,000), PegInsert (2,500), NutThread (2,500). Dual 384×384 cameras; per-frame `observation.state` (8-D), `action` (7-D), task-specific `physics_gt.*` channels including the corrected `contact_force` (bodyfilters approach). Per-episode physics parameters (mass, friction, damping, restitution where applicable) live in `meta/episodes.jsonl`.

Backbone and feature pools. V-JEPA 2 ViT-L (24 layers, 1024-d, 256-px input). Features extracted with `forward_resid_pre` monkey-patch (raw residual stream, no final LayerNorm) over 16-frame stride-1 windows. Per layer per window the raw $[2048, 1024]$ tensor is pooled into:

- *Variant A*: spatiotemporal mean over all 2048 tokens → 1024-d.
- *Variant B*: spatial mean within each of 8 temporal slots, flattened → 8192-d.

A subset of analyses additionally derive Variant C-like representations by slicing Variant B per slot.

Targets. Per-window targets keyed on the window's last frame index. Computed from `physics_gt.*` parquet columns. Acceleration uses unified central finite difference for all six tasks. Direction targets are masked NaN where $\|\mathbf{v}\| < 10^{-4}$. Quaternions are canonicalized ($q_w \geq 0$). Force magnitudes use `log1p` to handle heavy tails. Target tiers (per task):

- Kinematic: `ee_position`, `ee_velocity`, `ee_acceleration`, `ee_speed`, `ee_direction`, `obj_position`, `obj_velocity`, `obj_acceleration`, `obj_speed`, `obj_direction`, `ee_to_object_distance` / `ee_to_target_distance`, `phase`.
- Rotational: `ee_orientation (quat)`, `ee_angular_velocity`, `obj_orientation`, `obj_angular_velocity`.
- Contact: `contact_flag`, `contact_force`, `contact_force_mag`, `contact_force_log1p_mag`, `contact_point`.
- Progress (task-specific): `insertion_depth`, `peg_hole_lateral_error` (PegInsert); `axial_progress` (NutThread); `drawer_joint_pos`, `drawer_joint_vel`, `drawer_opening_extent` (Drawer); `ball_planar_travel_distance` (Strike).

Probe. Window-level Ridge with 5-fold `GroupKFold` on episode_id (group disjointness asserted). Within-fold standardization using inner-train statistics only; two-pass stable variance with a 1% median std floor for high-D variants. For Variant A, the spec-mandated 20-config Adam sweep (LR$\in$ {1e-4, 3e-4, 1e-3, 3e-3, 5e-3} $\times$ WD$\in$ {0.01, 0.1, 0.4, 0.8}, 100 epochs, batch 1024, MSE loss, variance-weighted R^2). Cosine similarity additionally reported for direction targets. For high-D analyses (Variant B 8192-d, temporal-order ablation) we use a GPU-accelerated closed-form Ridge ($\alpha = 1$) with CPU sklearn fallback on OOM, on 20K-row subsamples.

Integrity gates (12a–12d).

- 12a: shape and pooling-identity check (Variant A vs temporal-mean of Variant B). Tolerance: relative diff $< 10^{-3}$ in fp16 (necessary because raw activation magnitudes ~ 250).
- 12b: target consistency (velocity $\approx$ finite-diff position; speed $= \|\mathbf{v}\|$; direction-NaN where speed $< 10^{-4}$).
- 12c: 5-fold GroupKFold disjointness on episode_id.
- 12d: negative control — shuffle Push targets at episode level, run L12 ee_velocity Variant A through full sweep, require mean $R^2 < 0.05$.

All four gates passed on the recollected dataset (logs in `probe/results/logs/`).

Outputs

- Variant A per-target CSVs: `probe/results/<task>/variant_A/<target>.csv` (cols: layer, fold, best_lr, best_wd, r2, mse, n_test_windows, cos_sim_mean for direction).
- Variant A summary: `probe/results/<task>/variant_A/_summary.csv`.
- Variant A peak-layer table: `probe/results/peak_layers.csv`.
- Variant A decision: `probe/results/decision.json`; narrative: `probe/results/REPORT.md`.
- Variant B parallel files under `variant_B/`.
- Cache (mean-pooled): `/mnt/md1/solee/physprobe_features/<task>/variant_{A,B}/episode_<id>.npz` with `cache/manifest.json`.

Key observations

- Decision rule (`decision.json`): strict verdict `MARGINAL`, relaxed (99%-of-max plateau) verdict `HEALTHY`. Push `ee_velocity` argmax sits at L22; the first layer reaching 99% of that plateau is L17 (inside the 6–18 PEZ band).
- Best-layer R² for the spec anchors:
  Push `ee_velocity` L22 R² $= 0.951$,
  Push `ee_position` L22 R² $= 0.984$,
  Strike `ee_velocity` L22 R² $= 0.962$.
- Phase target trivially saturates (R^2=1.000 at L0) for Drawer, PegInsert, NutThread — the dataset stamps a constant phase code (always 7 for RL policies; constant for many scripted rollouts), so this target carries no real signal.
- Contact-flag for PegInsert and NutThread also saturates at L0 (R^2=1.000); these tasks are always-in-contact, so the variable is constant.

Known limitations

- Strict argmax for several targets falls in L21–L23, just outside the spec's 6–18 PEZ band; the relaxed plateau criterion is what keeps the verdict `HEALTHY`.
- Variant B has documented per-fold instability for direction targets at certain layers (high-D Ridge variance); affected rows are excluded in summary aggregates.
- Drawer randomizes only `drawer_joint_damping`; handle friction/mass are fixed (Isaac Lab env limitation).
- Contact-flag and phase targets are constant for some tasks; do not interpret L0 saturation as "information emerges at the input."

Status

Valid. Two pipelines (Variant A and Variant B) successfully completed and pushed to `Leesangoh/PhysREPA_Tasks`.

Note 4 — Contact and force probing

Setup

Variant A and Variant B Ridge probes for the contact tier (`contact_flag`, `contact_force` (3-D vec), `contact_force_mag` (1-D, raw N), `contact_force_log1p_mag` (1-D, log1p N), `contact_point` (3-D, defined where in-contact)). Same probe protocol as Section . Phase-/contact-conditional re-probing (Phase F) splits windows by `contact_flag` or by discrete `phase` value, and re-runs the Ridge probe per condition.

Outputs

- `probe/results/<task>/variant_{A,B}/contact_*.csv`
- Phase F: `probe/trajectory_analysis_B/results/stats/phase_conditional.csv` (2,136 rows: 6 tasks $\times$ 24 layers $\times$ 4 targets $\times$ 1–5 conditions).
- Phase F plots: `probe/trajectory_analysis_B/results/plots/phase_conditional_<task>_<target>.png` (24 panels).

Key observations (Variant A best-layer R²)

- Push: `contact_flag` L21 R²$= 0.768$; `contact_force_log1p_mag` L21 R²$= 0.754$; `contact_force` (3-D) L22 R²$= 0.345$.
- Strike: `contact_flag` L19 R²$= 0.825$; `contact_force_log1p_mag` L21 R²$= 0.834$; `contact_force` L21 R²$= 0.568$.
- Drawer: `contact_flag` L17 R²$= 0.710$; `contact_force_log1p_mag` L18 R²$= 0.720$; `contact_force_mag` L13 R²$= 0.752$.
- PegInsert: `contact_force_log1p_mag` L21 R²$= 0.474$ (contact_flag is constant, see limitations); `contact_point` L8 R²$= 0.846$.
- NutThread: contact targets weak; `contact_force_log1p_mag` L18 R²$= 0.079$.
- Reach: contact channels not present (negative control).

Phase-conditional probing (Variant A, contact-varying tasks).

- Drawer `contact_force_log1p_mag`: contact=0 best L0 R²$= -0.06$; contact=1 best L23 R²$= 0.83$. Force signal is only present during contact (expected).
- Push `ee_velocity`: unconditional L21 R²$= 0.926$; contact=0 L21 R²$= 0.935$; contact=1 L22 R²$= 0.849$. Kinematics are robust to contact state.
- Strike same pattern: unconditional L20 R²$= 0.941$; both contact slices stay above 0.9.
- Reach: unconditional L19 R²$= -0.097$ — negative-control check passes (no contact, kinematics dominated by random movements with little Reach-internal correlation).

Partial RSA (contact, controlling for pos+vel). After regressing out pos and vel, residual contact RSA is near zero for push, strike, drawer, nut_thread, and reach (max $|r| < 0.22$ across all layers). PegInsert is the only task with consistently positive residual contact RSA (≈ 0.20 at deep layers), suggesting jamming/rebound force structure that is not explained by EE motion alone.

F4-A — Physical-condition modulation of contact representations

Setup. For each randomized physics parameter, we partition episodes into Low/Med/High terciles using the episode-level parameter value from `episodes.jsonl`, then rerun the Variant A Ridge probe within each bin. We report the paired bootstrap CI of $\Delta R^2 = R^2_{\text{High}} - R^2_{\text{Low}}$ over the

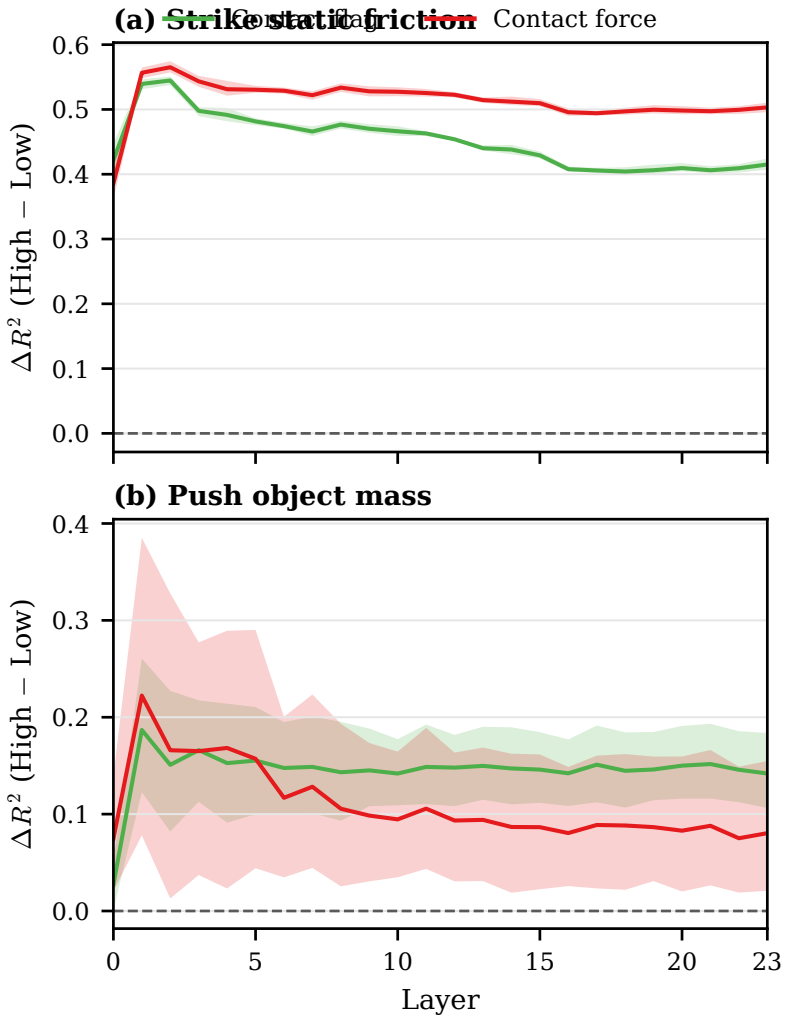

Figure 1: Physical-condition modulation of contact decodability. High-vs-low physics bins produce layer-wide gains on contact targets, with the strongest effect in Strike under static-friction split and a smaller but consistent effect in Push under object-mass split. Shaded regions show paired-bootstrap 95% CIs.

5 within-bin GroupKFold splits. By construction, this is not a "physics parameter probe" in the Phase C sense: the parameter itself remains a poor direct target, but it may still modulate how readable contact variables become.

Outputs.

- `probe/results/physics_condition_split/<task>__<param>.csv` (22 aggregated CSVs + per-fold sidecars).
- `probe/results/bootstrap_cis.csv` filtered to `claim=F4A_high_minus_low` (1,872 rows).

Strike static friction is the largest layer-wide modulation. The clearest effect appears in Strike when binning by `object_0_static_friction`. For `contact_force_log1p_mag`, the top layer is L2 with $\Delta R^2 = +0.565$ and a tight 95% CI $[+0.557, +0.574]$ ($p < 0.001$). The companion `contact_flag` effect is nearly identical: L2 $\Delta R^2 = +0.545$ with CI $[+0.537, +0.551]$. More importantly, the effect is not localized to a single peak layer: across roughly L1–L9, both contact

targets stay in the $+0.53$ to $+0.57$ range. The physical interpretation is straightforward: higher friction produces more sustained and predictable contact traces, which the representation makes easier to linearly decode.

Push object mass shows a smaller but consistent gain. Push exhibits the same phenomenon at lower magnitude when split by `object_0_mass`. For `contact_force_log1p_mag`, the best layer is L1 with $\Delta R^2 = +0.222$ and CI $[+0.078, +0.385]$; for `contact_flag`, the best layer is again L1 with $\Delta R^2 = +0.187$ and CI $[+0.123, +0.261]$. The gains are not confined to early layers: `contact_flag` stays near $+0.15$ across mid/deep layers (e.g. L17/L20/L21), indicating that the mass effect persists across the same representational depth where the base contact probe already peaks. Heavier objects plausibly amplify the force consequences of a given impact/drag velocity, making contact state more readable.

Drawer joint damping is a noisy counterexample. Drawer does not provide the same confirmatory signal. The largest `ee_acceleration` gap appears at L3 with $\Delta R^2 = +2.46$, but its CI is extremely wide ($[+0.009, +7.33]$), and neighboring layers are small or unstable. Under the project's exclusion rules, this is treated as a noisy counterexample rather than paper-level positive evidence.

General pattern and paper-level nuance. Two points matter. First, the modulation is *layer-wide*, not just a change in the single best PEZ layer. Second, on contact targets its magnitude can exceed the Phase F5 frame-shuffle effect on the same targets: F5 contact deltas were only about $+0.05$ at peak, whereas Strike static-friction modulation reaches about $+0.55$. Combined with Phase C's negative result, this yields the key nuance of the paper: *V-JEPA does not directly linearly encode physics parameters as targets, yet it strongly reflects their downstream effect on contact variable decodability.*

Known limitations

- Several tasks have constant contact channels (PegInsert and NutThread are always in contact $\Rightarrow$ `contact_flag` is the constant 1; Reach has no contact channel). The L0 $R^2 = 1$ for PegInsert/NutThread `contact_flag` is the trivial constant-prediction; ignore in cross-task comparisons.
- `contact_point` is undefined where `contact_flag`=0; we mask NaN. Variance is dominated by spatial position.
- Variant B contact probing on certain tasks shows fold instability — compare against Variant A as anchor.
- Extremely large F4-A deltas at unstable layers (e.g. Strike L0/L1 under object-mass split) are not interpreted as real signal when the base foldwise R^2 is deeply negative or unstable.

Status

Valid for the recollected dataset (contact GT verified). Phase-conditional re-probing and physical-condition split probing are complete. Cross-model comparison is now available in Appendix Note H.

Note 5 — Latent-trajectory geometry

Setup

A battery of geometric and structural analyses applied to V-JEPA 2 ViT-L window features per layer, per task. Sampling: 30–60 episodes per task, window-stride aligned. Where applicable, both Variant A (1024-d) and Variant B (8192-d) are computed.

Analyses run.

- *Trajectory stats*: per-window path length, max/mean speed (in latent), curvature, direct distance, direct/path ratio.

- *Intrinsic dimension*: PCA-90/95/99 components per (task, layer); top-1 / top-10 explained variance ratios.
- *Cross-layer CKA*: 24×24 layerwise similarity per task.
- *Cross-task CKA*: pairwise 6×6 task similarity per layer; layer-averaged trend.
- *Partial RSA*: target RDM correlation after regressing out pos+vel; targets = acc, contact.
- *Tangent RSA*: latent-velocity RDM vs physics-velocity RDM (`rsa_vel_tan`); same for acceleration with pos+vel partialed (`rsa_acc_tan_partial`).
- *Event-locked*: latent speed in a window of $\tau \in [-8, +8]$ frames around contact onset.
- *Kinematic-subspace CCA*: per-layer 5-component CCA between latent and physics groups {pos, vel, acc, contact}.
- *Koopman self-predictability*: linear 1-step latent forecaster $z_{t+1} \approx Az_t$; report R^2_{self} and gain over persistence baseline (ΔR^2).

Outputs

- `probe/trajectory_analysis_B/results/stats/`: `trajectory_stats.csv`, `trajectory_stats_summary.csv`, `intrinsic_dim.csv`, `cross_layer_cka_<task>.csv`, `cross_task_cka_layerwise.csv`, `cross_task_cka_avg_per_layer.csv`, `rsa_partial.csv`, `rsa_tangent.csv`, `event_locked.csv`, `cca_subspace.csv`, `koopman.csv`, `speed_profiles.csv`, `per_slot_*.csv` (Phase B).
- Plots: `probe/trajectory_analysis_B/results/plots/` (62 PNGs across the analysis suite).

Key observations

Path length and curvature. Latent path length grows roughly monotonically with depth across all tasks (e.g. Strike: L0≈326 → L23≈2,888; Drawer: L0≈544 → L23≈5,011). Curvature saturates near ∼1.9–2.0 from L4 onward. Direct/path ratio is ∼0.01–0.07 at every depth ⇒ trajectories are highly curved; almost no Euclidean-straight motion.

Cross-task CKA. Average across 15 task-pairs per layer peaks at L3 (0.903), then monotonically falls to L23 (0.357). L0 lower (0.527) than L1–L3. ⇒ even in self-supervised pretraining, deep layers specialize per task.

Kinematic-subspace CCA. `rho1` per group, averaged across tasks:

Layer	pos	vel	acc	ct
0	0.945	0.638	0.419	0.734
8	0.988	0.850	0.556	0.839
12	0.993	0.868	0.653	0.861
16	0.994	0.889	0.674	0.866
20	0.996	0.892	0.683	0.856
23	0.996	0.879	0.648	0.872

Position is essentially perfectly aligned at all depths; velocity and contact saturate by L8; acceleration alone keeps refining through L20.

Tangent RSA (latent vs physics velocity). `rsa_vel_tan` per task at deep layers shows a non-trivial non-monotonic trace for Push and Strike:

Layer	push	strike	drawer	reach	peg_insert	nut_thread
0	+0.46	+0.72	+0.83	+0.79	+0.13	+0.04
8	−0.24	−0.08	+0.52	+0.71	+0.15	+0.08
12	−0.15	+0.32	+0.61	+0.81	+0.19	+0.13
16	+0.16	+0.51	+0.69	+0.84	+0.17	+0.12
20	+0.23	+0.58	+0.68	+0.84	+0.17	+0.12
23	+0.22	+0.57	+0.68	+0.81	+0.15	+0.10

Push and Strike show a sign reversal in the L4–L12 range. Reach (the no-contact control) is uniformly strong and stable.

Koopman self-predictability gain. ΔR^2 over persistence baseline decreases with depth for push/strike/peg_insert/nut_thread/reach. Drawer is the outlier ($\Delta R^2 \in [0.36, 0.50]$ across all depths).

Layer	push	strike	drawer	reach	peg	nut
0	$+0.21$	$+0.39$	$+0.50$	$+0.16$	$+0.12$	$+0.09$
8	$+0.21$	$+0.31$	$+0.46$	$+0.23$	$+0.15$	$+0.17$
16	$+0.16$	$+0.20$	$+0.38$	$+0.18$	$+0.13$	$+0.16$
23	$+0.07$	$+0.09$	$+0.36$	$+0.02$	$+0.07$	$+0.11$

$\Rightarrow$ deeper layers carry more information but are less linearly predictable from their own past.

Per-temporal-slot decomposition (Phase B, Variant B). Slot 7 (window's last frame) consistently has lower intrinsic dimension and higher per-slot probe R² than slots 0–6:

slot	0	1	2	3	4	5	6	7
PCA-90 (avg)	26.2	28.8	22.8	30.5	30.5	30.5	28.6	22.1
PCA-90 @ L20	30.8	36.5	25.3	40.3	40.2	37.3	34.3	20.2
`ee_velocity` R² (L18–22)	0.156	0.166	0.185	0.209	0.235	0.292	0.332	0.434

Partial RSA. After controlling for pos+vel, residual acceleration RSA stays in $[0.10, 0.43]$ for drawer/strike/reach, smaller (0.10–0.19) for peg/nut, and 0.00–0.25 for push. Contact RSA is near zero except for peg_insert (≈ 0.20).

F1-b — Phase-space geometry of latent trajectories

Setup. For each task and layer, we form a latent phase-space trajectory from window-aligned features z_t and first differences $\Delta z_t = z_{t+1} - z_t$. Geometric diagnostics are computed from 30 sampled episodes per task (360 rows total in the final CSV): mean latent speed $\|\Delta z\|$, phase-space curvature κ, sweep volume in the $(\mathrm{PC1}(z), \mathrm{PC1}(\Delta z))$ plane, and a local Lyapunov-style statistic. For Push / Strike / Drawer, the same measurements are also split by the three-way contact partition used in F2 (`pre`, `during`, `post`).

Outputs. `probe/trajectory_analysis_B/results/stats/phase_space.csv` (360 rows; task, layer, split, and 5 geometry statistics) and `probe/trajectory_analysis_B/results/plots/phase_space_<task>.png` (layerwise mean speed / curvature / sweep-volume panels).

Key observations.

- **Mean latent speed rises strongly with depth across all six tasks.** On the all-window split, Push grows from $\|\Delta z\| = 0.172$ at L0 to 2.806 at L23; Drawer from 0.225 (L0) to 2.724 (L23); NutThread from 0.116 (L0) to 2.966 (L23). Reach is the one mild exception: it peaks slightly earlier at L8 (3.231) and then stays high through late layers.
- **Curvature collapses from very high early values to much smoother mid/late trajectories.** Push drops from $\kappa = 17.14$ at L0 to 1.41 at L23; Strike from 26.88 at L0 to 1.06 by L8; Reach from 18.23 at L0 to 0.57 at L8. The consistent pattern is "fast but jagged" early geometry transitioning into a smoother latent flow after the first few layers.
- **Contact-rich windows occupy a larger and faster latent regime than pre-contact windows.** For Push, the split-average latent speed is 1.603 (`pre`) vs 1.775 (`during`), while the split-average sweep volume expands from 21.7 (`pre`) to 69.1 (`during`) and 70.6 (`post`). Strike shows the same pattern: speed $1.651 \rightarrow 1.813$ and sweep volume $16.3 \rightarrow 70.5$ from `pre` to `during`, with `post` staying even broader (94.4).
- **The largest phase-space expansion often happens in shallow-to-mid layers, not at the deepest layer.** The all-window sweep-volume maximum occurs at L8 for Push (166.2), L8 for Strike (186.0), and L8 for Reach (175.3), while late layers keep higher $\|\Delta z\|$ but occupy

a narrower 2D phase-space footprint. This is consistent with a "mid-layer expansion, late-layer compression" geometry.

- **Drawer behaves as a long-contact outlier.** Under the current contact-phase split, Drawer has almost no usable `pre` support; the informative comparison is `during` vs `post`. During-contact curvature remains elevated (3.06 vs 2.90 on the all-window split), while the deepest layers still reach high latent speeds (peak 2.74 at L23), consistent with prolonged contact-rich manipulation rather than a single sharp impact.

Status. Complete supplementary geometric diagnostic. The phase-space results are kept as supporting evidence for the representation-hierarchy story rather than promoted to a main-text causal claim in the way F5 is.

Known limitations

- All trajectory analyses are descriptive; no significance tests or null distributions are computed yet.
- Episode subsampling is fixed seed (42), 30–60 eps/task; the seed-stability of every observation is not yet measured.
- Variant B per-slot intrinsic-dim values for layers with extreme outliers may be capped by PCA component count.

Status

Valid. All 17 stats CSVs and 62 plots produced and pushed.

Note 6 — Cross-task transfer and temporal-order ablation

Setup

Phase E — cross-task transfer (Variant A). For ordered task pairs (src, tgt), train a Ridge probe on `src` task's features and labels, evaluate on `tgt`. Standardization uses `src` statistics (transfer-style; tgt features are not seen at training time). Compare *transfer R²* to within-task 5-fold GroupKFold R² on `tgt`; report *gap* = within − transfer.

Pairs: {push↔strike, push↔drawer, strike↔drawer} (6 directed pairs). **Targets:** `ee_velocity, ee_acceleration, contact_flag, contact_force_log1p_mag`. **Layers:** all 24. **Episodes:** 50 per task.

Phase D — temporal-order ablation (Variant B). Variant B feature is $[N, 24, 8192]$ with the 8192-d axis split into 8 slots of 1024-d each. For each scheme, transform the slot-axis, re-flatten, and run a closed-form Ridge probe (GPU `torch.linalg.solve` with sklearn CPU fallback on OOM, α=1).

Schemes: {baseline_full, mean_over_slots, single_slot_k ($k\in\{0,3,7\}$), prefix_k ($k\in\{0,3,7\}$), reverse, random_global_perm (one fixed permutation across the batch), random_per_sample (fresh permutation per window)}. **Targets:** `ee_acceleration, obj_acceleration, contact_flag, contact_force_log1p_mag`. **Tasks:** push, strike, drawer. **Layers probed:** 12 layers (0, 2, 5, 8, 11, 13, 15, 17, 19, 20, 22, 23). **Subsample:** 20,000 rows.

Outputs

- `probe/trajectory_analysis_B/results/stats/cross_task_transfer.csv` (576 rows: 6 pairs × 24 layers × 4 targets).
- `probe/trajectory_analysis_B/results/stats/cross_task_transfer_per_fold.csv` (fold-level transfer rows for bootstrap confirmation).
- `probe/results/bootstrap_cis.csv` filtered to `claim=transfer_gap` (576 rows).
- `probe/trajectory_analysis_B/results/stats/temporal_order_ablation.csv` (1,584 rows).

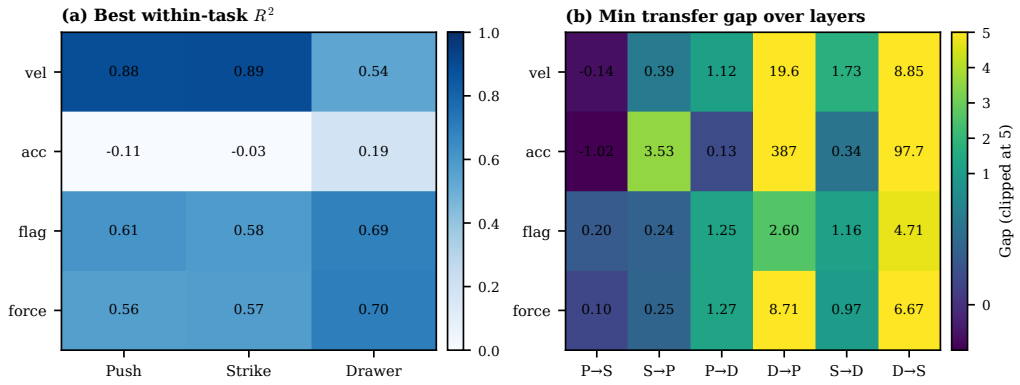

Figure 2: Cross-task structure summarized at the best achievable layer per cell. Left: best within-task R^2 by task and target. Right: minimum transfer gap (within-task minus cross-task) over layers; color is clipped at 5 while cell annotations show the true gap. Push and Strike retain a small-gap regime, whereas Drawer remains a clear outlier.

- Plots: `cross_task_transfer_*.png` (8) and `temporal_order_<task>_<target>.png` (12).

Key observations

Cross-task transfer (best layer per pair, `ee_velocity`).

src → tgt	best L	transfer R²	within tgt R²
push → strike	17	+0.746	+0.885
strike → push	18	+0.495	+0.880
push → drawer	11	−0.747	+0.376
strike → drawer	14	−1.302	+0.428
drawer → push	15	−18.7	+0.833
drawer → strike	15	−8.0	+0.836

Push and Strike share a transferable kinematic subspace; Drawer is on a disjoint manifold (transfer collapses, within-task remains strong).

Bootstrap update (transfer-gap confidence). We additionally bootstrap the *gap* = within-target − transfer over the 5 within-target folds (576 CI rows total). The pattern is sharp: **Drawer-directed transfer is consistently broken**, whereas **Push/Strike is the only pair with a small-gap regime**. Representative examples:

pair / target / layer	gap	95% CI	note
push → strike, `ee_velocity`, L4	+0.045	[−0.023, +0.112]	CI crosses 0
push → strike, `ee_acc`, L22	−0.026	[−0.158, +0.107]	CI crosses 0
push → strike, `contact_force`, L7	+0.097	[−0.113, +0.307]	CI crosses 0
push → drawer, `ee_velocity`, L11	+1.123	[+1.077, +1.177]	CI > 0
strike → drawer, `ee_acc`, L17	+0.336	[+0.135, +0.546]	CI > 0
drawer → push, `contact_flag`, L19	+2.604	[+2.562, +2.648]	CI > 0

At the pair level, Push→Strike is the only direction with multiple layers whose CIs include zero for dynamics/contact-force targets (`ee_velocity`: 3/24 layers; `ee_acc`: 3/24; `contact_force_log1p_mag`: 5/24). In contrast, every Drawer↔(Push/Strike) pairing has CI-excluding-zero gaps at essentially all layers, often by large margins.

Interpretation. Cross-task transfer and Phase F5 answer complementary questions. F5 asks whether a representation relies on *within-task temporal coherence*; transfer asks whether that representation is *task-general once the task changes*. Together they point to the same conclusion:

Push and Strike share a partially transferable physics subspace, but Drawer remains strongly task-entangled. In short, the model learns some task-general kinematics, but not a fully task-agnostic geometry of manipulation.

Temporal-order ablation (best-baseline-layer R^2, per task/target).

target / task	baseline	reverse	rand_global	rand_per_sample
push `ee_acc`	-0.965	-0.961	-0.961	-1.420
strike `ee_acc`	$+0.280$	$+0.280$	$+0.281$	-0.125
drawer `ee_acc`	$+0.108$	$+0.109$	$+0.109$	-0.870
push `contact_flag`	$+0.231$	$+0.233$	$+0.232$	$+0.282$
strike `contact_flag`	$+0.051$	$+0.047$	$+0.047$	-0.044
drawer `contact_flag`	$+0.411$	$+0.420$	$+0.408$	-0.415
push `contact_force`	-0.296	-0.304	-0.306	-0.224
strike `contact_force`	-0.253	-0.260	-0.257	-0.446
drawer `contact_force`	$+0.591$	$+0.591$	$+0.583$	-0.741

Mean-over-slots vs baseline. Collapsing the slot axis (i.e. Variant A) sometimes *outperforms* Variant B baseline:

- Push `contact_flag`: 0.458 (mean) vs 0.231 (baseline).
- Strike `contact_flag`: 0.144 vs 0.051.
- Push `contact_force`: $+0.236$ vs -0.296.

But Drawer `contact_force` flips: -0.114 (mean) vs $+0.591$ (baseline). Drawer is the only task where per-slot temporal layout contributes meaningfully to Ridge fit.

Reverse $\approx$ random_global_perm $\approx$ baseline. Across all 36 (task $\times$ target) cells, reverse and global-permutation schemes match the baseline within $|\Delta R^2| < 0.02$. Only `random_per_sample` — the scheme that breaks cross-sample slot identity — causes a clear drop. $\Rightarrow$ a linear probe on Variant B exploits per-slot *content*, not temporal *order*; consistent permutations are absorbed by Ridge weights.

Known limitations

- Phase D and E only sweep {push, strike, drawer}. PegInsert / NutThread / Reach not covered.
- Negative R^2 values for several baseline cells reflect Ridge underfit at 8192-d with 20K subsample; absolute magnitudes should not be over-interpreted, only the relative ordering across schemes.
- Drawer's outlier behavior (separate manifold, per-slot helps, Koopman gain is high) is consistent across analyses but may partly reflect dataset asymmetry (only `joint_damping` randomized).

F5 — input-side frame-shuffle

Setup. Phase D's slot-level temporal-order ablation showed that a linear probe on Variant B does not exploit slot order: *reverse* $\approx$ *global permutation* $\approx$ *baseline* for nearly every (task, target). This narrows but does not answer the upstream question: does V-JEPA 2's forward pass itself use frame order? F5 addresses this by shuffling the 16-frame input stack *per window* (fresh seed-controlled permutation, deterministic per (episode_id, shuffle_seed)) *before* the V-JEPA forward, then re-extracting Variant A features into `cache/<task>/variant_A_shuffled/episode_<id>.npz` and re-running the same probe protocol. **Targets:** `ee_velocity`, `ee_acceleration`, `contact_flag`, `contact_force_log1p_mag`. **Tasks:** push, strike (drawer deferred).

Pre-launch sanity (verified 2026-05-01). On a 3-episode dry-run, layer-0 Variant-A mean differs from the unshuffled extraction by max-abs $\approx 10^{-3}$ (frame order has minimal effect on patch-embed), while layer-23 mean differs by max-abs $\approx 1.5 \times 10^{-2}$ with Pearson correlation 0.19. The

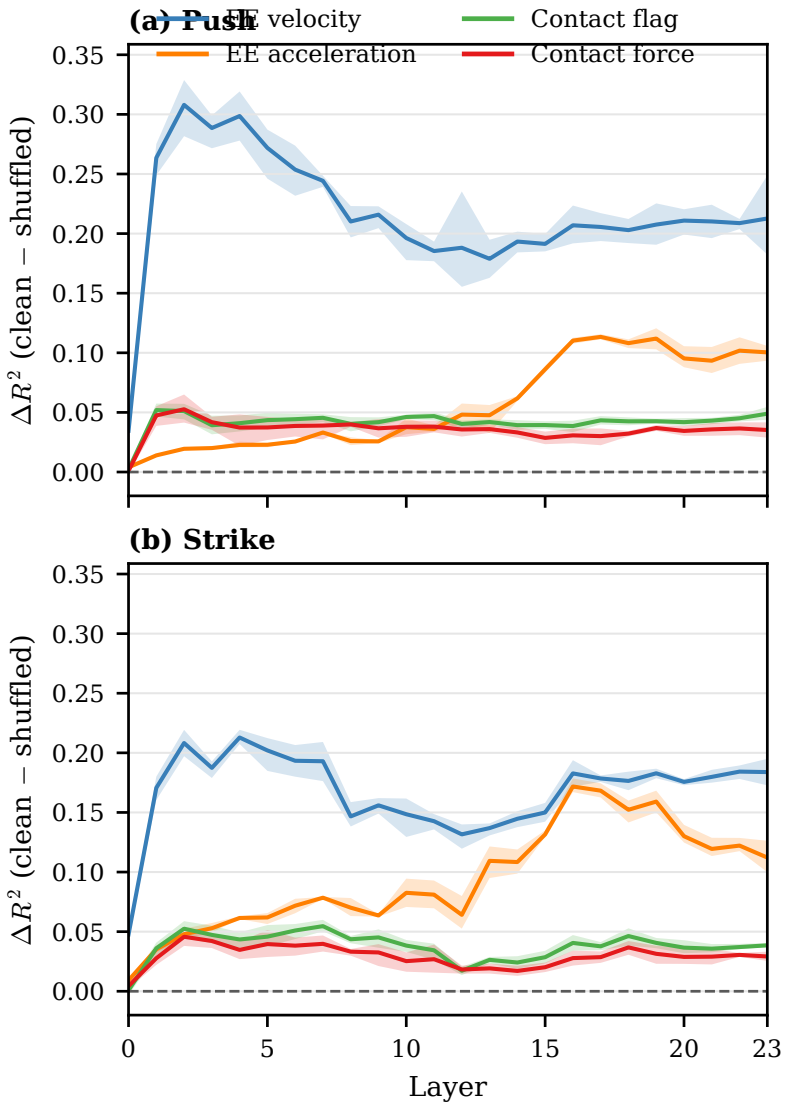

Figure 3: Frame-shuffle reveals target-selective temporal-coherence dependence. In both Push and Strike, motion targets lose substantially more R^2 than contact targets when the 16-frame input stack is shuffled before the V-JEPA forward. Curves show $\Delta R^2 = R^2_{\text{clean}} - R^2_{\text{shuffled}}$ with paired-bootstrap 95% CIs.

depth-dependent divergence confirms the implementation propagates input shuffling through the V-JEPA forward as intended; effect is small at the input and grows with depth.

Metric. Per (task, target, layer): *paired* bootstrap CI of $\Delta R^2 = R^2_{\text{unshuffled}} - R^2_{\text{shuffled}}$ over 5 GroupKFold folds, episode-level resampling, 1000 replicates, percentile. Schema in `probe/results/bootstrap_cis.csv` with `claim=F5_delta_r2`.

Observed pattern (final).

- Dynamics targets (`ee_velocity`, `ee_acceleration`) show large positive ΔR^2 and replicate across both Push and Strike.
- Contact targets (`contact_flag`, `contact_force_log1p_mag`) remain positive but much smaller, typically around $+0.05$ at peak.

- The hierarchy is stable across tasks: first-order motion peaks early, higher-order dynamics peak in the PEZ band, and late contact decoding is comparatively order-robust.

The headline reads: "frame-level temporal coherence is selectively necessary for dynamic targets and only weakly necessary for late contact decoding." Combined with the Phase D slot-level result, the joint statement is: "input temporal coherence matters; slot order is not linearly exploited by the probe."

Outputs (when complete).

- `probe/results/<task>/variant_A_shuffled/<target>.csv`: per-fold rows, schema identical to the main probe.
- `probe/results/bootstrap_cis.csv`: rows with `claim=F5_delta_r2`, `task` $\in$ {push, strike}, `target` $\in$ {4 above}, layer $\in \{0..23\}$.

Push results (final, 2026-05-01 14:48). Probe sweep on shuffled features completed in ~ 30 min wallclock (Adam 20-HP grid is GPU-efficient on 1024-d features — well below Codex's conservative 4 h estimate). Bootstrap-paired CIs (1000 reps, percentile, episode-level fold pairing):

Target	Peak ΔR^2	Peak Layer	95% CI	CI excludes 0
`ee_velocity`	$+0.308$	L2	$[+0.282, +0.329]$	24 / 24
`ee_acceleration`	$+0.113$	L17	$[+0.111, +0.116]$	24 / 24
`contact_force_log1p_mag`	$+0.053$	L2	$[+0.041, +0.065]$	23 / 24
`contact_flag`	$+0.052$	L1	$[+0.047, +0.057]$	24 / 24

Three-tier hierarchy (the paper's central mechanistic story). Push exhibits a clean depth-structured organization of frame-coherence dependence:

- **Early (L1–L2): first-order motion.** `ee_velocity` peaks at L2 with $\Delta R^2 = +0.308$. The shuffle damage is largest precisely where short-range frame correspondence is used; once the model can no longer match consecutive frame identities, instantaneous velocity is the first thing to collapse.
- **Mid (L17, PEZ band): higher-order dynamics.** `ee_acceleration` peaks at L17 with $\Delta R^2 = +0.113$. Acceleration cannot be read off a single frame difference; it requires longer-range temporal integration, so its dependence on coherent ordering is concentrated at the depth where that integration happens.
- **Late (L20+): order-robust contact state.** `contact_flag` and `contact_force_log1p_mag` show their largest shuffle sensitivity at L1–L2 and decay to near zero by L20. The main probe's high contact decodability at L21 is therefore not contradicted: the late representation is a *configuration / state* encoding that recovers contact from instantaneous spatial cues, not from temporally ordered motion.

Codex / paper-writer framing (per RECALL discussion 14:48). "Temporal coherence is universally useful, but its necessity is strongly target- and depth-dependent. Low-order motion depends on early temporal coherence; higher-order dynamics peak in the PEZ band; late contact decoding is comparatively order-robust." We position F5 as the paper's *killer mechanistic evidence*: it turns a phenomenological emergence map into a causal-style intervention result.

Strike replication (final, 2026-05-01 17:39). F5 strike extraction completed (3000 eps in 383.5 min); shuffled probe auto-launched via `watch_strike` on all 4 GPUs in parallel and finished in ~ 11 min. Final paired bootstrap CIs:

Target	Peak ΔR^2	Peak Layer	95% CI	CI excludes 0
`ee_velocity`	$+0.213$	L4	$[+0.207, +0.220]$	24 / 24
`ee_acceleration`	$+0.172$	L16	$[+0.167, +0.177]$	24 / 24
`contact_force_log1p_mag`	$+0.046$	L2	$[+0.038, +0.054]$	24 / 24
`contact_flag`	$+0.055$	L7	$[+0.050, +0.060]$	23 / 24

The hierarchy is preserved exactly: dynamic targets show large positive ΔR^2 (acc/vel $\gg$ contact). Two task-specific differences from push: **(i)** strike's `ee_acceleration` peak is *larger* than push's (+0.172 vs +0.113) — consistent with strike being an impulsive impact task with sharper acceleration transients that the model encodes in the PEZ band; **(ii)** strike's `ee_velocity` peak is slightly later (L4 vs L2) and a bit smaller in magnitude (+0.213 vs +0.308), reflecting the difference between sustained sliding (push) and impulsive contact (strike) in how velocity is exposed.

This cross-task replication is the paper's primary robustness check on the F5 mechanism: the qualitative pattern survives task change while the magnitudes shift in the direction predicted by the underlying physics. Combined with R3M's failure on the same dynamic targets (Note H), F5 establishes the V-JEPA representation's specifically temporal-coherence-dependent character.

Status (final F5, 2026-05-01 17:39). Push and strike F5 shuffled probes complete. Bootstrap CI rerun yields `F5_delta_r2` = 192 rows (push 96 + strike 96) in `bootstrap_cis.csv`. R3M baseline (Note H) covers push, strike, drawer. This note should now be read together with Figure 3 (causal-style within-task intervention) and Figure 2 (cross-task structure probe).

Status

Phase E/D/F5 are complete and integrated. Transfer, temporal-order ablation, and frame-shuffle now form a single evidence block for task-general vs task-entangled temporal structure.

Note 7 — Discussion

1. What is now established. The corrected per-window pipeline now supports a coherent main story. First, V-JEPA 2 ViT-L exhibits a depth-structured emergence profile: kinematic, contact, and progress variables peak at different layers, and the PEZ band is most strongly associated with higher-order dynamics. Second, the frame-shuffle intervention (Phase F5) converts that phenomenology into a causal-style result: motion targets collapse far more than contact targets when input temporal coherence is destroyed. Third, physical-condition splits (Phase F4-A) show that contact decodability is not fixed; it is systematically modulated by latent physics regime. Fourth, the R3M baseline (Appendix Note H) establishes that these effects are not generic to a strong image encoder.

2. What F5 resolves about temporal coherence. The central mechanistic claim is now sharper than in the original revision plan. *Temporal coherence is universally useful, but its necessity is strongly target- and depth-dependent.* In both Push and Strike, `ee_velocity` suffers the largest early-layer frame-shuffle loss, while `ee_acceleration` peaks later in the mid-PEZ band. Contact targets are different: their shuffle penalty is small, concentrated early, and largely disappears at late layers. This supports a three-stage hierarchy: early layers expose first-order motion, mid layers integrate higher-order dynamics, and late layers encode a more order-robust contact-as-state representation.

3. What F4-A adds beyond F5. F5 shows that coherent ordering matters for dynamics; F4-A shows that the *same representation* is also sensitive to latent physics regime. The key nuance is that *physics parameters are not directly linearly decodable as targets, yet they strongly modulate the decodability of contact variables.* This is the clearest in Strike, where `object_0_static_friction` changes `contact_flag/contact_force_log1p_mag` by about $+0.55$ R^2 across L1–L9, and still visible in Push, where `object_0_mass` produces a smaller but consistent $+0.15$–$+0.22$ gain. The modulation is often larger than the F5 contact shuffle effect itself, which means that the representation is not simply reading out contact/no-contact snapshots; it is reflecting the downstream consequences of task physics on those contact traces.

4. Why Drawer is the critical outlier. Drawer is no longer a nuisance exception; it is now one of the most informative tasks in the suite. Transfer collapses between Drawer and the Push/Strike pair, making it the clearest example of task entanglement in the current representation. The R3M baseline also fails most sharply on Drawer, with the largest gap on `ee_acceleration` and contact force. These findings agree with the task's mechanics: Drawer is a slow, constrained, articulated 1-DOF process whose relevant state changes are weakly recoverable from a single image but become

readable under temporal pretraining. In other words, Drawer is the strongest evidence that V-JEPA's advantage is not limited to impact-like motion; it also extends to sustained articulated dynamics.

5. Remaining open questions. Three questions remain substantial enough to motivate the next paper pass. First, the current mainline evidence is V-JEPA-specific; the post-reset VideoMAE-L and DINOv2-L reruns are still needed before any objective-dissociation claim is re-asserted. Second, scale remains open: ViT-G may shift PEZ depth, F5 sensitivity, or transfer structure, but the new pipeline has not yet been run end-to-end at that scale. Third, the action-control side is still incomplete: the previous OOD action-chunk result was invalidated, and a clean reproduction is still required before any control-layer claim can be trusted. At a finer grain, Drawer also remains a useful stress test for disentangling representation facts from dataset asymmetry, since only `joint_damping` currently varies.

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Note A — Artifact manifest

Repository

`Leesangoh/PhysREPA_Tasks` on GitHub. All probing artifacts under `probe/`. Plan 3+ recovery commits: `4643f83` (Phases B/C/E/F) and `8d2ba68` (Phase D rerun).

Code paths

- Variant A/B feature extraction: `probe/scripts/01_extract_features.py`.
- Targets builder: `probe/scripts/02_build_targets.py`, `probe/utils/targets.py`.
- Probe runner: `probe/scripts/03_run_probe.py` (`--tier` flag controls target subset).
- Aggregation / decision: `probe/scripts/04_aggregate_results.py`, `05_write_report.py`.
- Episode-level physics: `probe/scripts/07_episode_physics_probe.py`.
- Trajectory analysis (Variant A): scripts under `probe/trajectory_analysis/`.
- Trajectory analysis (Variant B): scripts 01–13 under `probe/trajectory_analysis_B/scripts/`; Phase B/D/E/F scripts 14–17 in the same directory.

Data paths

- Dataset: `/home/solee/data/data/isaac_physrepa_v2_recollected_2026-04-23/`.
- Feature cache: `/mnt/md1/solee/physprobe_features/<task>/variant_{A,B}/episode_<id>.npz`.
- V-JEPA 2 ViT-L weights: `/mnt/md1/solee/checkpoints/vjepa2/vitl.pt`.

Result CSVs (post-reset pipeline)

- Per-task per-target Variant A: `probe/results/<task>/variant_A/<target>.csv`.
- Per-task per-target Variant B: `probe/results/<task>/variant_B/<target>.csv`.
- Variant A summaries: `probe/results/<task>/variant_A/_summary.csv`.
- Decision: `probe/results/decision.json`; narrative: `probe/results/REPORT.md`.
- Peak-layer table: `probe/results/peak_layers.csv`.
- Trajectory-analysis stats (17 CSVs): `probe/trajectory_analysis_B/results/stats/`.
- Plots (62 + 49): `probe/trajectory_analysis_B/results/plots/` and `probe/results/plots/`.

Note B — Run config

Backbone. V-JEPA 2 ViT-L, 24 layers, 1024-d, 256-px input, raw residual stream (`forward_resid_pre` monkey-patch). Tubelet=2 $\Rightarrow$ 16 input frames produce 8 temporal tokens; spatial grid 16$\times$16=256 $\Rightarrow$ 2048 tokens/window total.

Probe (Variant A, primary recipe). 5-fold `GroupKFold` on `episode_id`. Within each fold: 10% inner-train$\rightarrow$inner-val episode-level split. Within-fold z-score using inner-train statistics; two-pass stable variance with 1% median std floor. 20-config Adam grid (LR$\in$ {1e-4, 3e-4, 1e-3, 3e-3, 5e-3} $\times$ WD$\in$ {0.01, 0.1, 0.4, 0.8}), 100 epochs, batch 1024, MSE loss. Best HP picked by inner-val MSE; predict test; unnormalize; report `r2_score(..., multioutput='variance_weighted')` plus per-fold MSE and (for direction) cosine similarity.

Probe (Variant B and trajectory closures). Closed-form Ridge (α=1) via `torch.linalg.solve` on GPU; sklearn CPU `Ridge` fallback on CUDA OOM. fp16 GPU storage with index-only slicing for memory efficiency.

Subsampling for high-D analyses. Phase D temporal-order ablation: 20,000 rows per (task, layer, target, scheme). Phase E cross-task transfer: 50 episodes per task. Phase F phase-conditional: 60 episodes per task. Trajectory analyses (CKA, RSA, Koopman): 30–60 episodes per task with seed 42.

Integrity gates passed (12a–12d). Pooling-identity max relative diff $< 10^{-3}$; target-consistency checks pass; GroupKFold disjointness asserted; negative control ($R^2 < 0.05$ on shuffled targets) pass.

Note C — Episode-level physics-parameter probing

Setup

`probe/scripts/07_episode_physics_probe.py`. Per task, per layer, per physics-parameter: stack episode-mean features (Variant A) of all sampled episodes $\rightarrow X \in \mathbb{R}^{n_{\text{eps}} \times 1024}$; target Y is the per-episode physics value from `meta/episodes.jsonl`. 5-fold KFold over episodes; standardize within fold; sklearn Ridge (α=1); report R^2 mean/std. Episode budget: ≤ 200 per task (typically ~ 196 after NaN filtering).

Outputs

`probe/results/<task>/physics_params/_summary.csv` and `_per_fold.csv`; plots `probe/results/plots/physics_param_<task>_A.png`.

Key observations (best-layer R² per param)

task	param	best L	R² mean ± std
push	object_0_dynamic_friction	8	-0.484 ± 0.317
push	object_0_mass	20	-0.734 ± 0.162
push	surface_static_friction	20	-0.681 ± 0.210
strike	object_0_static_friction	15	$+0.090 \pm 0.267$
strike	object_0_restitution	7	-0.702 ± 0.551
drawer	drawer_joint_damping	3	-0.752 ± 0.219
peg_insert	peg_mass	19	-0.784 ± 0.228
peg_insert	hole_static_friction	10	-0.820 ± 0.266
nut_thread	nut_mass	13	-0.878 ± 0.161
nut_thread	bolt_static_friction	6	-1.020 ± 0.538

Reading. Negative R^2 at every (task, param, layer) means episode-mean V-JEPA features do not linearly recover physics parameters at the episode level. The single positive value (Strike static friction at L15, $+0.09$) is within the inner-fold noise band.

Known limitations

- Episode budget is small (~ 196) relative to feature dimension (1024). Ridge is statistically underpowered.
- Episode-mean pooling discards within-episode variation; physics params may be encoded only in event-windows. A windowed formulation that aggregates features only during contact / impact remains untried.
- Variant B (8192-d) episode-level probing was deferred for the same statistical-power reason and is not in this artifact set.

Status

Valid as a negative-result note; not a final claim.

Note D — Exclusions and gotchas

- Ignore `contact_flag` and `phase` R^2 for tasks where these channels are constant: PegInsert and NutThread (always-in-contact ⇒ flag=1); Drawer/PegInsert/ NutThread `phase=7` always.
- `contact_point` probing R^2 is dominated by spatial position when contact occurs; treat as a position proxy.
- Strike `contact_force_log1p_mag` is more comparable across tasks than the raw `contact_force_mag` because of heavy tails (max forces 5 N for peg, 429 N for drawer).
- Variant B per-fold variance for direction targets is large at certain layers; rely on Variant A as anchor for direction targets unless explicitly studying slot-dependence.
- Drawer randomization is single-parameter (`joint_damping`); contact-tier randomness is dataset-induced rather than parameter-induced. Apparent drawer outlier status may be partly a dataset artifact.

Note H — R3M (image encoder) context baseline

Setup

We ran a minimal R3M baseline on Push, Strike, and Drawer, using the robotics-pretrained `r3m` ResNet-50 checkpoint as a *static image encoder baseline* rather than a temporal competitor. Each MP4 was decoded framewise; every frame was resized from 384×384 to 224×224, ImageNet-normalized, and forwarded independently through the ResNet-50 backbone. We extracted five stage-wise feature streams (`stage_0` through `stage_4`), applied spatial global-average pooling per frame, then mean-pooled over the same 16-frame windows used everywhere else in the paper. Probing used the same 5-fold `GroupKFold` protocol and the same 20-HP Adam grid as V-JEPA Variant A.

Outputs

- Per-fold probe CSVs: `probe/results/<task>/r3m/{ee_acceleration, contact_flag, contact_force_log1p_mag}.csv`
- Stage summaries: `probe/results/<task>/r3m/_summary.csv`, `probe/results/<task>/r3m/_best_stage.csv`
- Side-by-side comparison against V-JEPA: `probe/results/r3m_vs_vjepa_summary.csv`

Key observations

V-JEPA beats R3M on all 9 matched cells.

task	target	R3M best	V-JEPA best	Δ (R3M−V-JEPA)
push	ee_acceleration	stage_4, 0.084	L19, 0.219	−0.135
push	contact_flag	stage_3, 0.677	L21, 0.768	−0.091
push	contact_force_log1p_mag	stage_3, 0.638	L21, 0.754	−0.116
strike	ee_acceleration	stage_4, 0.189	L17, 0.413	−0.224
strike	contact_flag	stage_3, 0.678	L19, 0.825	−0.147
strike	contact_force_log1p_mag	stage_3, 0.694	L21, 0.834	−0.140
drawer	ee_acceleration	stage_1, 0.004	L22, 0.298	−0.293
drawer	contact_flag	stage_4, 0.509	L17, 0.710	−0.201
drawer	contact_force_log1p_mag	stage_4, 0.425	L18, 0.720	−0.295

The largest deficit is on acceleration. The strongest gap appears on the most temporal target, `ee_acceleration`: Push loses by 0.135 R^2, Strike by 0.224 R^2, and Drawer by 0.293 R^2. Drawer is the sharpest case: R3M peaks at only 0.004 on `stage_1`, while later image stages turn negative (e.g. `stage_3` $= -0.352$, `stage_4` $= -0.141$). This is consistent with the main paper's interpretation that temporal coherence is especially important for higher-order dynamics, and that a static image encoder is poorly matched to that signal.

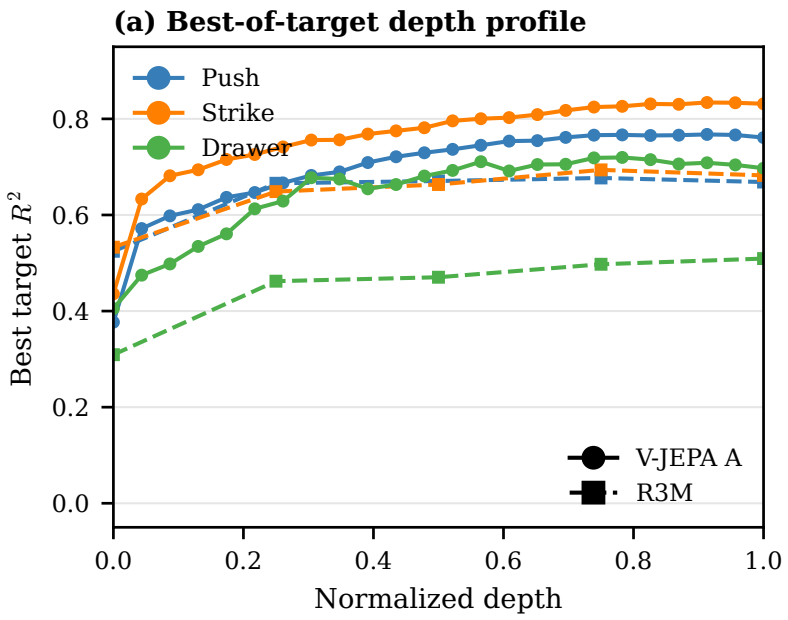

Figure 4: Depth-normalized comparison between R3M and V-JEPA. For each task and depth, the curve reports the best R^2 among {`ee_acceleration`, `contact_flag`, `contact_force_log1p_mag`}. R3M saturates by `stage_3/4`, while V-JEPA continues to improve into the deep PEZ band.

Contact is better, but still not enough. R3M is more competitive on the contact targets than on acceleration, but still loses robustly. On Push, the deficits are -0.091 (`contact_flag`) and -0.116 (`contact_force_log1p_mag`); on Strike they widen to -0.147 and -0.140, respectively; on Drawer they are larger still at -0.201 and -0.295. Thus even when contact geometry is partially visible in a single frame, image-only features do not close the gap to V-JEPA.

Drawer is the clearest temporal-pretraining advantage case. Among the three tasks, Drawer is the one where sustained articulated motion is most weakly recoverable from a single image. The drawer state changes gradually along a constrained 1-DOF trajectory, so `ee_acceleration` and `contact_force_log1p_mag` depend more on temporal integration than on snapshot appearance. This makes Drawer the cleanest task-level demonstration that V-JEPA's temporal pretraining advantage is not limited to impact-like motion (Strike), but also extends to slow, structured articulation.

R3M stage progression still saturates shallow. For both Push and Strike, the best contact stage is `stage_3`, while the best acceleration stage is `stage_4`. That is, R3M does show a coherent depth trend, but its useful hierarchy saturates within a 5-stage image backbone. By contrast, V-JEPA's best layers for the same targets occur deep in its 24-layer stack (L17–L21). Drawer reinforces the same point: even where R3M reaches its best contact scores at `stage_4`, it remains well below V-JEPA's late-layer peaks.

Status

Complete contextual baseline note. This comparison is kept in the appendix because R3M is an image-encoder context point, not a direct temporal-model competitor to V-JEPA.